Missing Data

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

Read the contract. Keep zero. Preserve unknowns and reasons.

Validate missing, typed and bounded values without inventing measurements or hiding excluded cases.

Keep one habit: Duration filter → separate stock contract → expanded reason audit → independent new bounds.

MAIN LESSON · SECTION 1

How can zero be valid while a missing value is not?

> Use the written type and range rules instead of a truthiness shortcut.

Your learning goals

I can detect absent and unusable values.

I can keep unknown distinct from zero.

I can report accepted and rejected records.

Recall text versus integer

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

- Is the text "12" the same type as the integer 12?

Your end goal

By the end, I can validate **missing value**s without confusing zero with absent data.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Noise and Outliers. Give an example and a mistake that the idea helps you avoid.

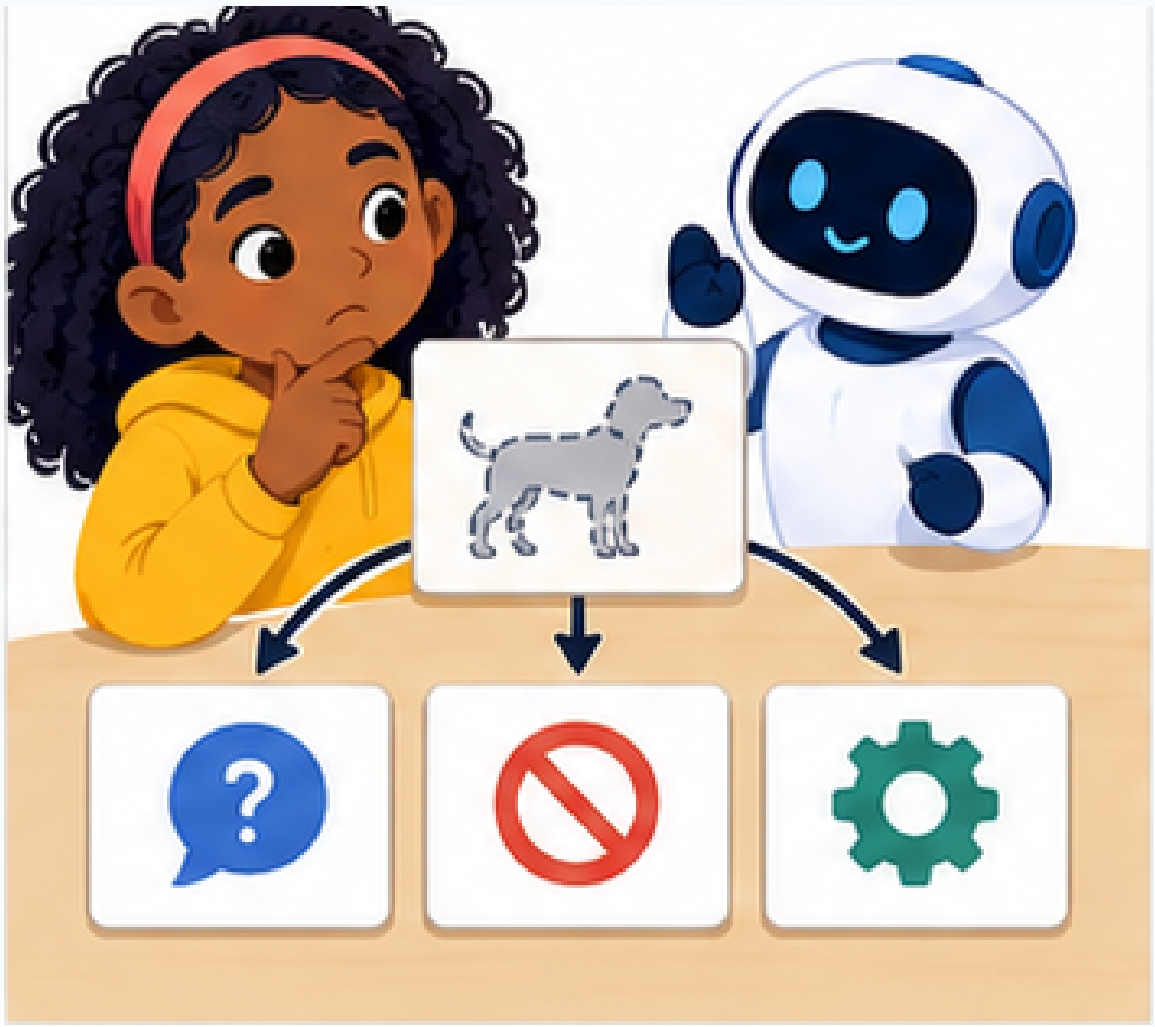
Without opening the old lesson, explain one key idea from Choose Informative Questions. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Accuracy as a Fraction. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 18: **Missing Data**

A system needs an explicit policy for unknown fields.
</details>



THINKER MISSION

Compare ask, **abstain**, and **default** strategies.

Thinker name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

This week’s end goal

MAIN LESSON · SECTION 2

State the duration contract before filtering

> Keep recorded zero, absent values and invalid values distinct.

The activity-log problem

A fictional activity log contains an absent duration, a recorded zero and an out-of-range value. Our schema accepts exact integer minutes from 0 through 60.

Keep rejected records visible for review instead of replacing their values with guesses.

Row meaning, units and review scope

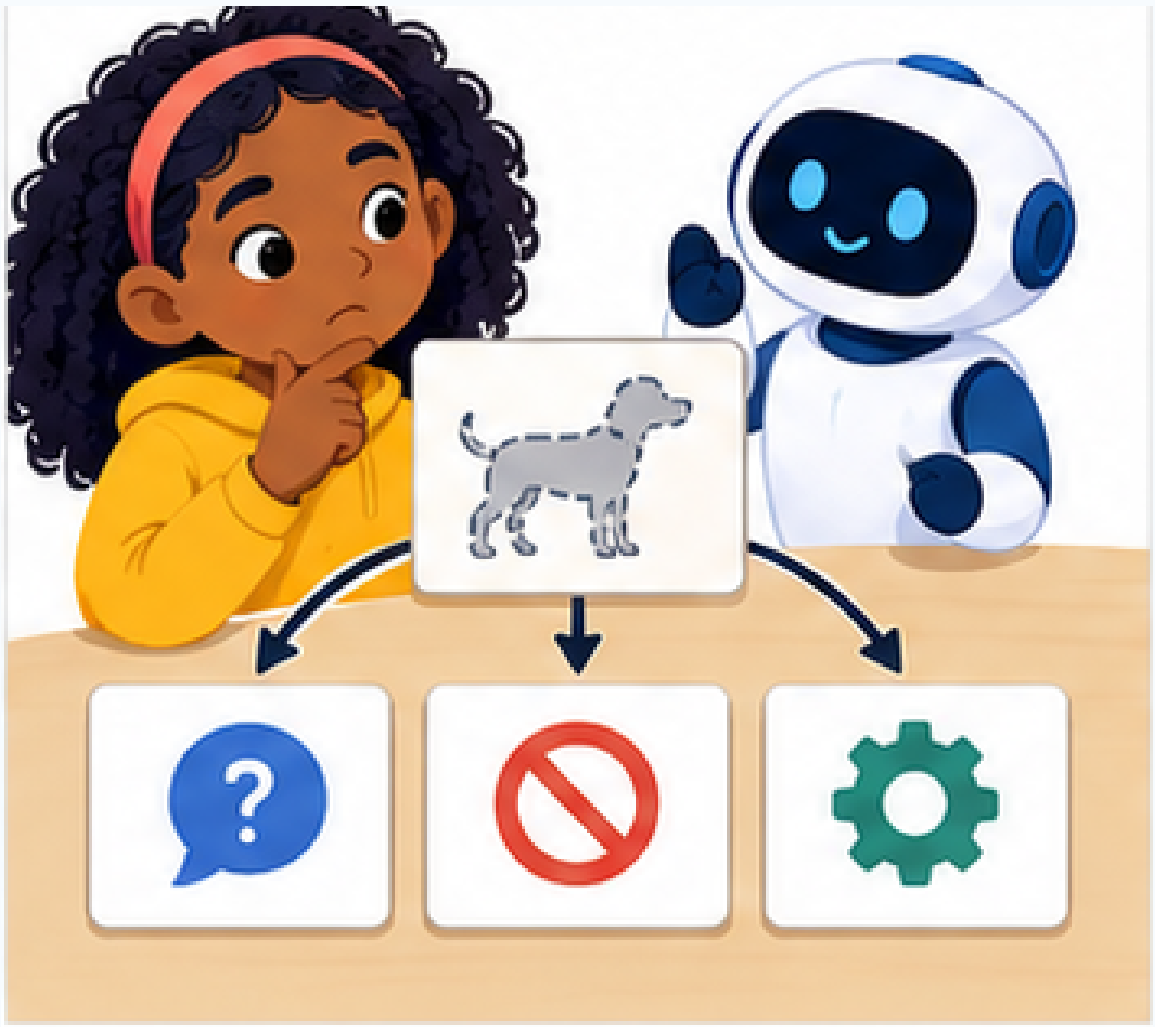
Each duration row is one fictional activity session. This contract checks only the minutes field: an exact integer 0–60 inclusive. A missing field or explicit None means unknown here. A type or range rejection means this version cannot use the record for its numeric summary; preserve the row for review. Acceptance does not prove that the measurement is true.

▼ **Earlier wording, alternatives and picture transcripts**

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use **missing data** to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

MAIN LESSON · SECTION 3

Trace the complete five-record filter

> Keep source order, accepted values and rejected count visible.

Predict and run the supplied script

```
rows = [{"minutes": 0}, {}, {"minutes": 12},
        {"minutes": -1}, {"minutes": None}]
accepted = []
rejected = 0
for row in rows:
    value = row.get("minutes")
    if type(value) is int and 0 <= value <= 60:
        accepted.append(value)
    else:
        rejected = rejected + 1
print(accepted)
print(rejected)
```

Explain get, append and the type-first condition

get reads the minutes field without raising an error when the key is absent. The word and short-circuits: Python checks the exact type first and only tests the numeric bounds if that check succeeds. Thus None and text never reach a numeric comparison. type(value) is int also rejects Boolean True and False; arithmetic compatibility is not the same as this declared input contract.

dict.get returns None when this key is absent. An explicit None also represents unknown in this dataset. append adds an item to the end of a list. The script prints [0, 12] and then 3.

The exact type check prevents comparing None or text with the numeric range.

Match each raw value to the contract

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
Recorded zero	0	Accept as a measurement
Absent or None	Unknown	Hold for review
Negative integer	Outside range	Reject under this schema

Read each row from left to right and connect the evidence to the stated rule.

A recorded 0 says the measured duration was zero. A missing field says we do not have the duration. They are different statements. Trace each record with three columns: raw value, decision and reason.

Keep 0 and 12; reject the absent value, -1 and None. The accepted list therefore has two values and the **rejection** count is three. Do not turn an unknown value into a number just to make the calculation run.

Point to the deciding check

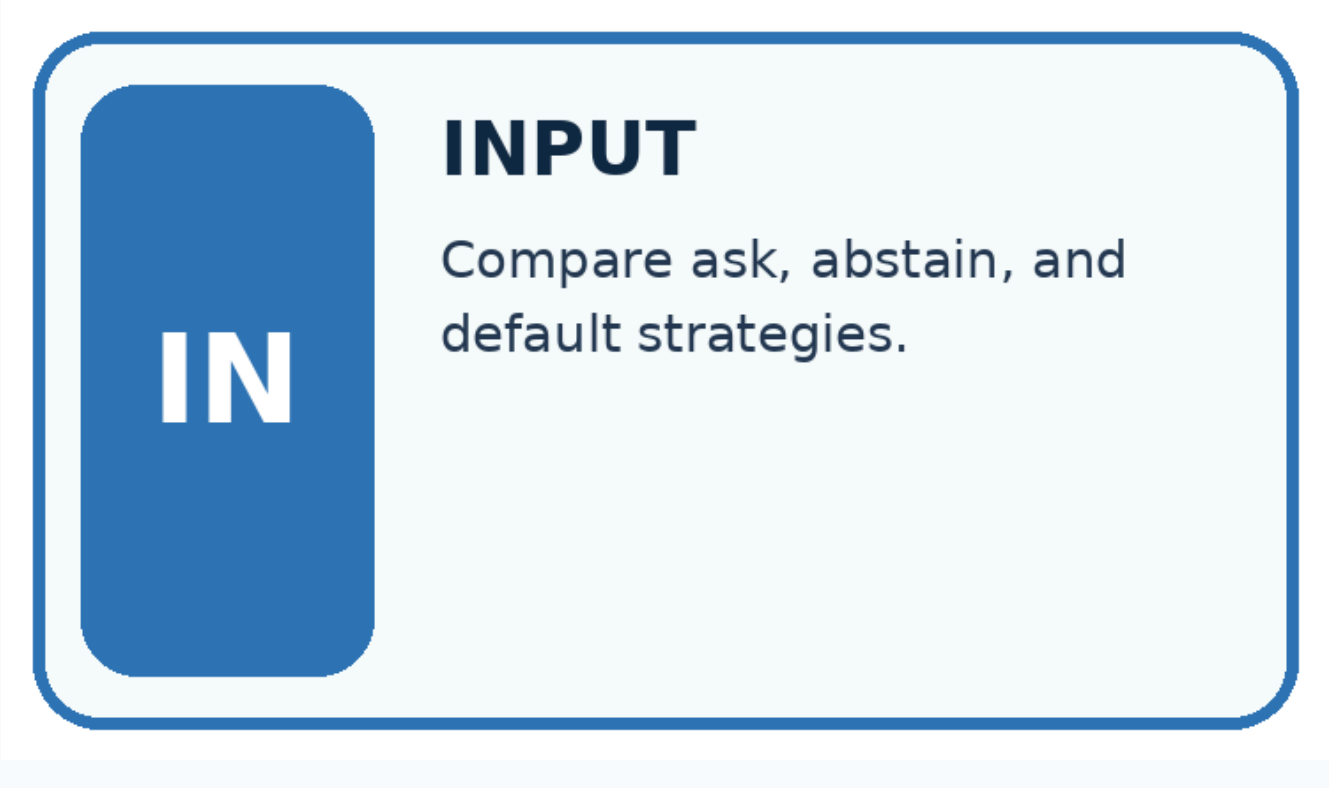
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

VISUAL **MECHANISM**

Input → **Process** → **Output**

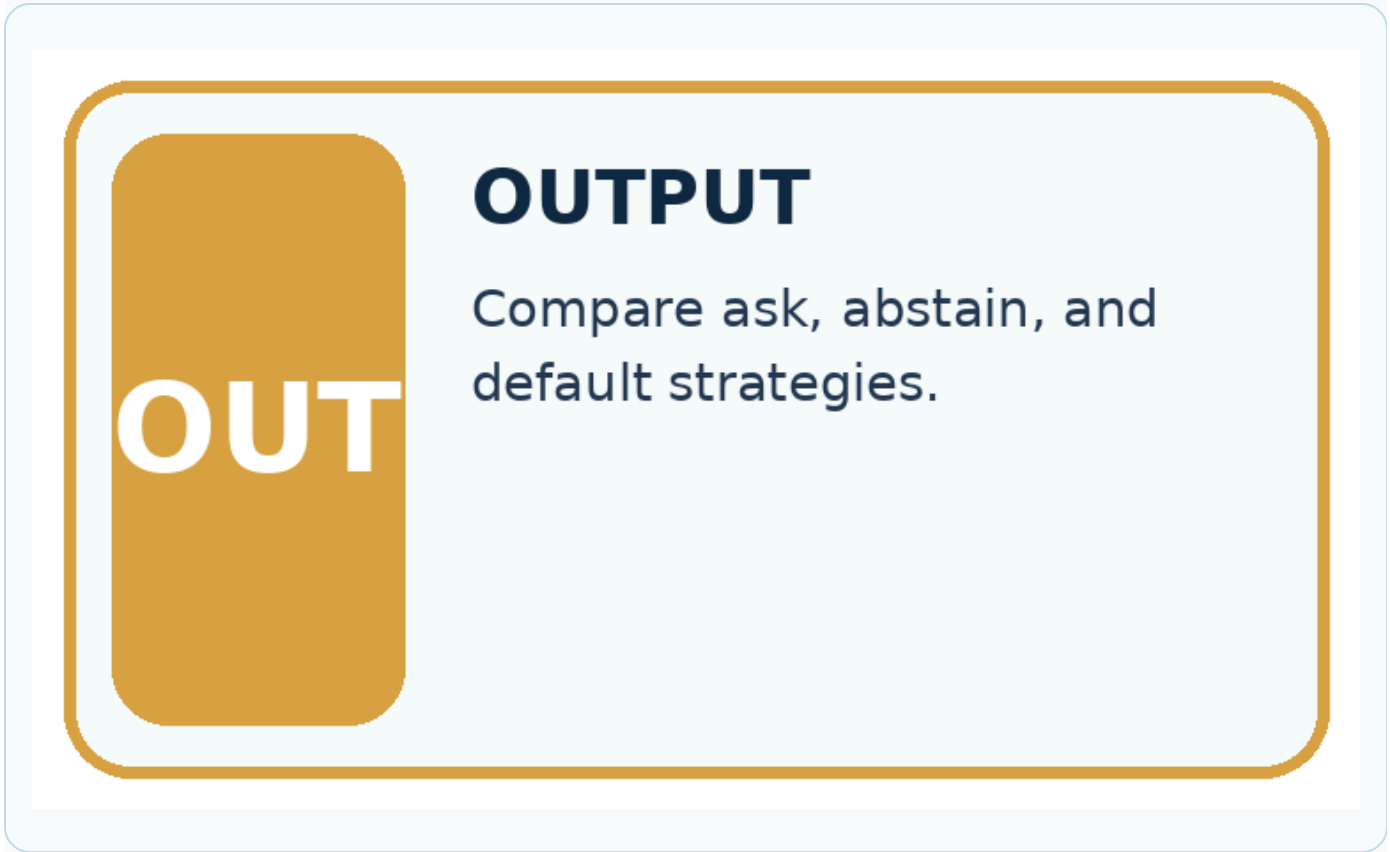
Trace the information. At each stage, ask what changed and how you could test it.



↓
</details>


PROCESS
A system needs an explicit policy for unknown fields.

↓


OUT
OUTPUT
Compare ask, abstain, and default strategies.

Which stage could fail even when the other two are correct? Explain:

> **Write or draw your reasoning and evidence.**

A closer explanation

Keep zero separate from unknown

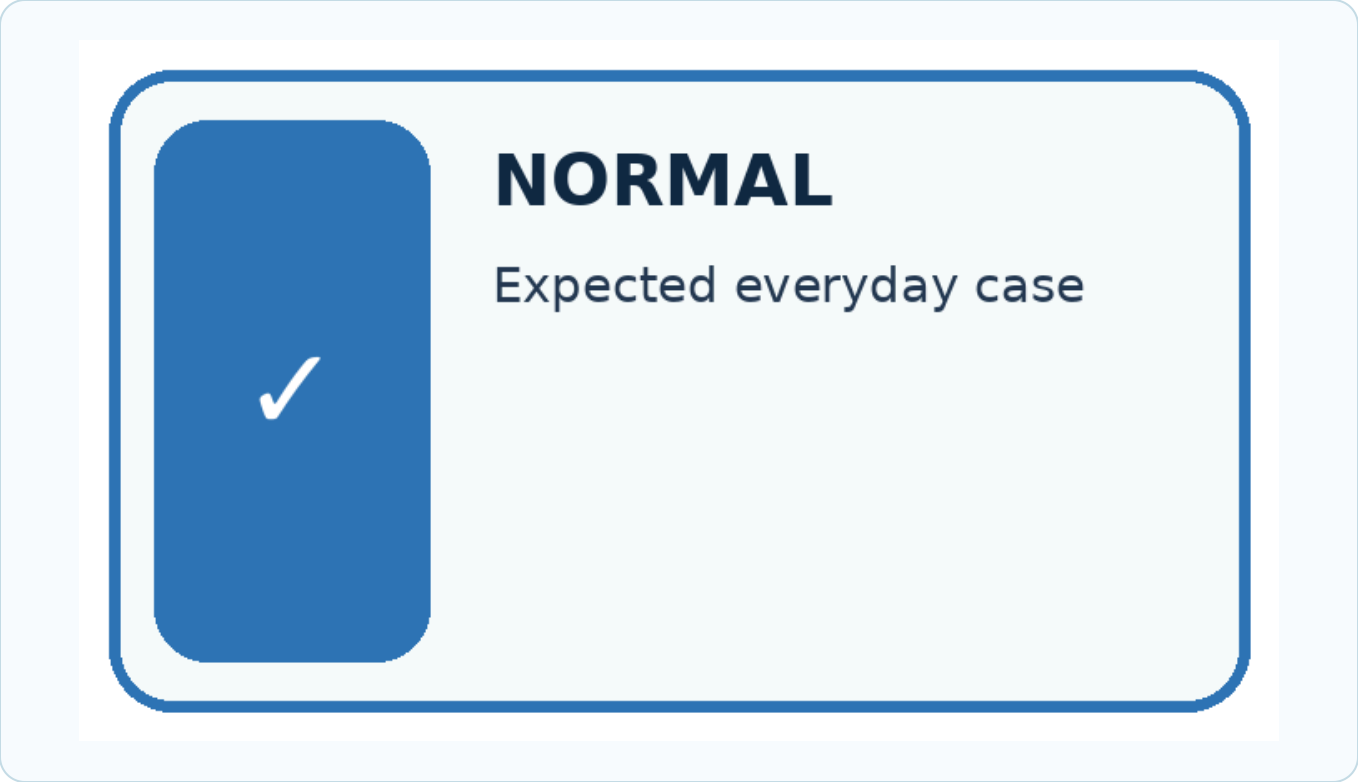
Connect the picture and the explanation

MAIN LESSON · SECTION 4

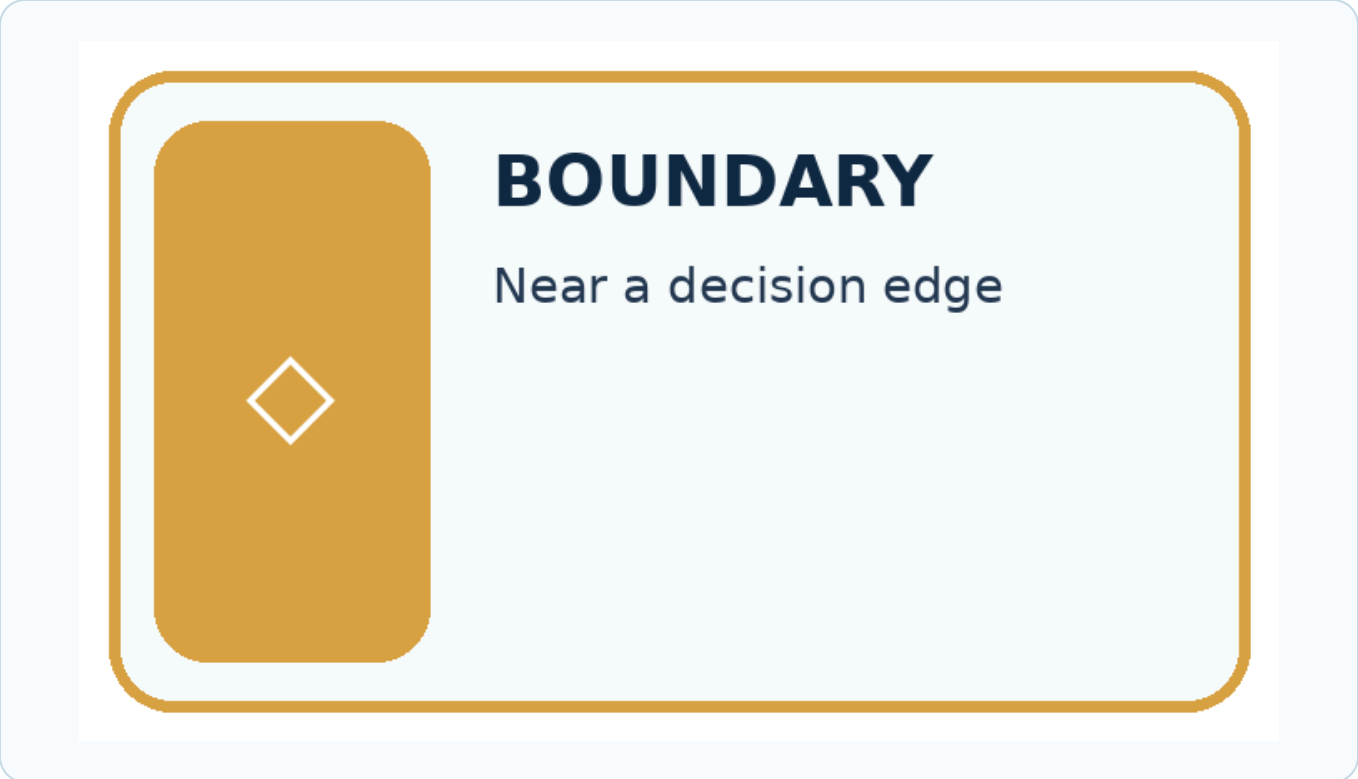
Compare a valid boundary with an unknown value

> Do not turn a rejected record into a guessed zero.

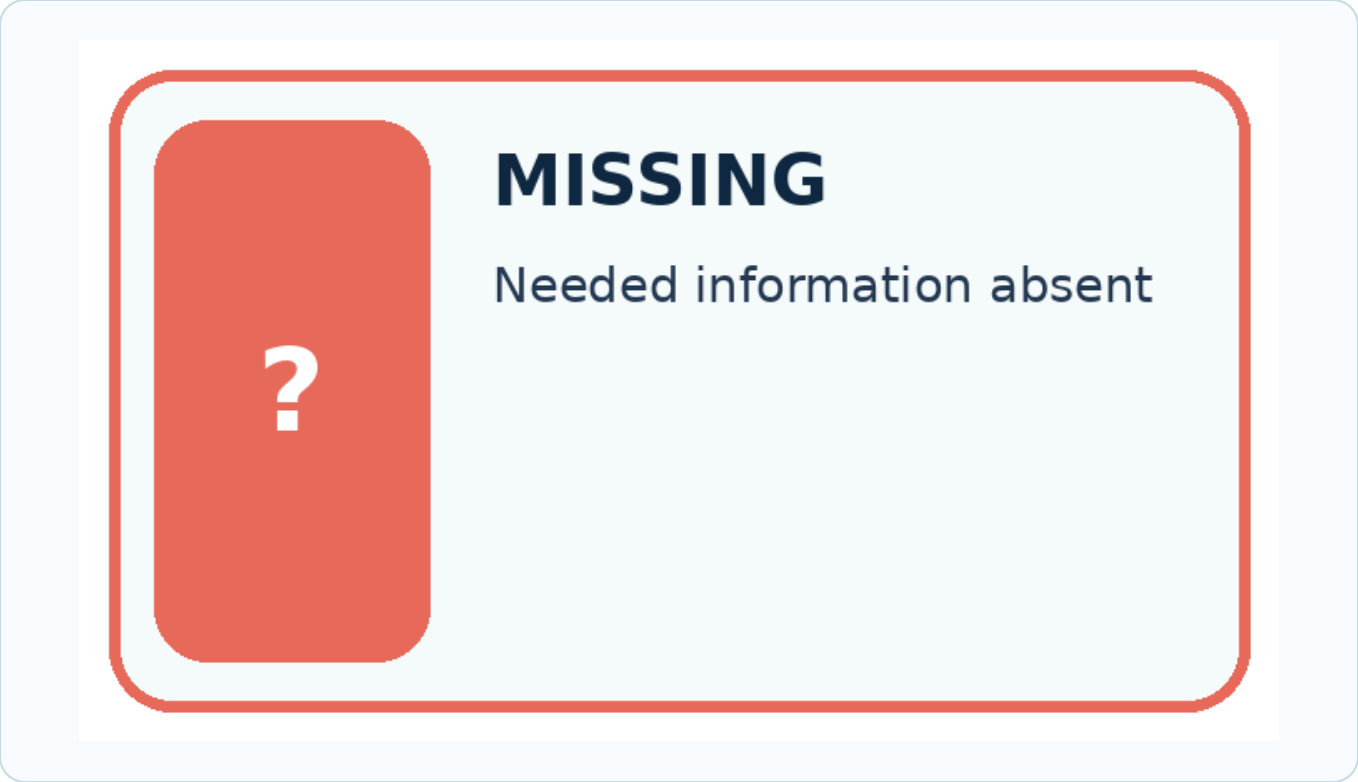
Four pictures with a purpose



Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

> **Write or draw your reasoning and evidence.**

Predict the supplied changed case

- Change: Replace text "4" with integer 4 through an authorised conversion.

▼ Check after predicting

Prediction: Accepted values become [0,4,4]; three rows remain rejected.

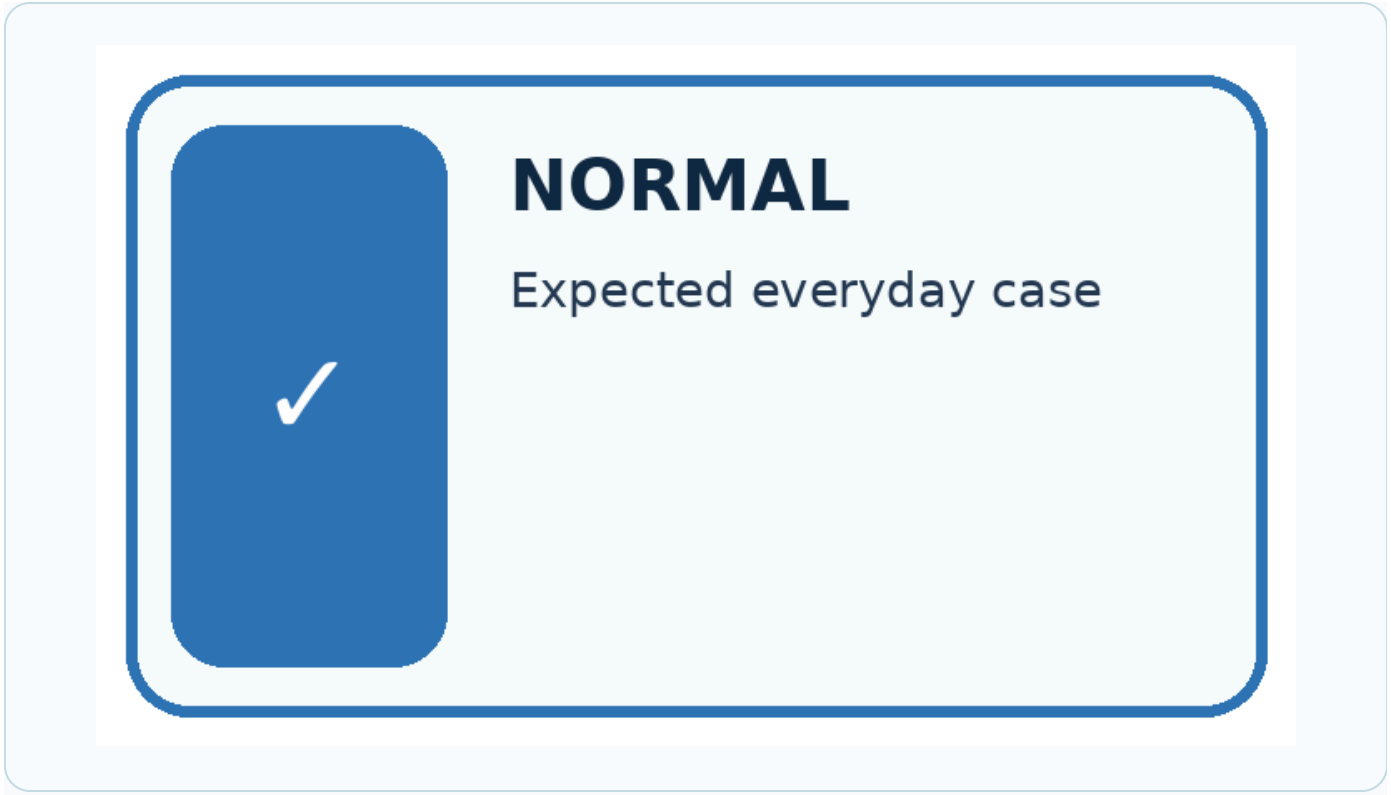
Reason: The converted value now satisfies both type and range checks.

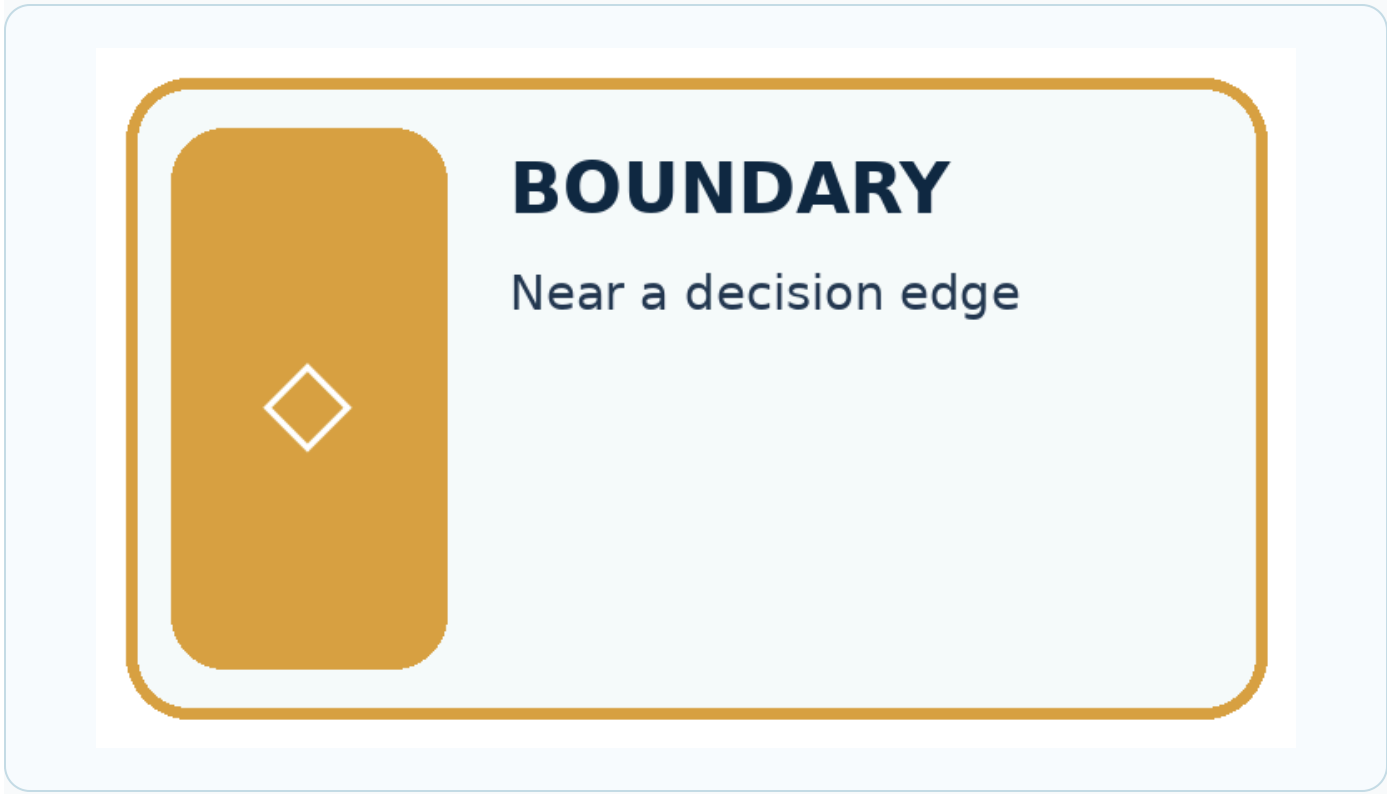
▼ Earlier wording, alternatives and picture transcripts

CASE GALLERY

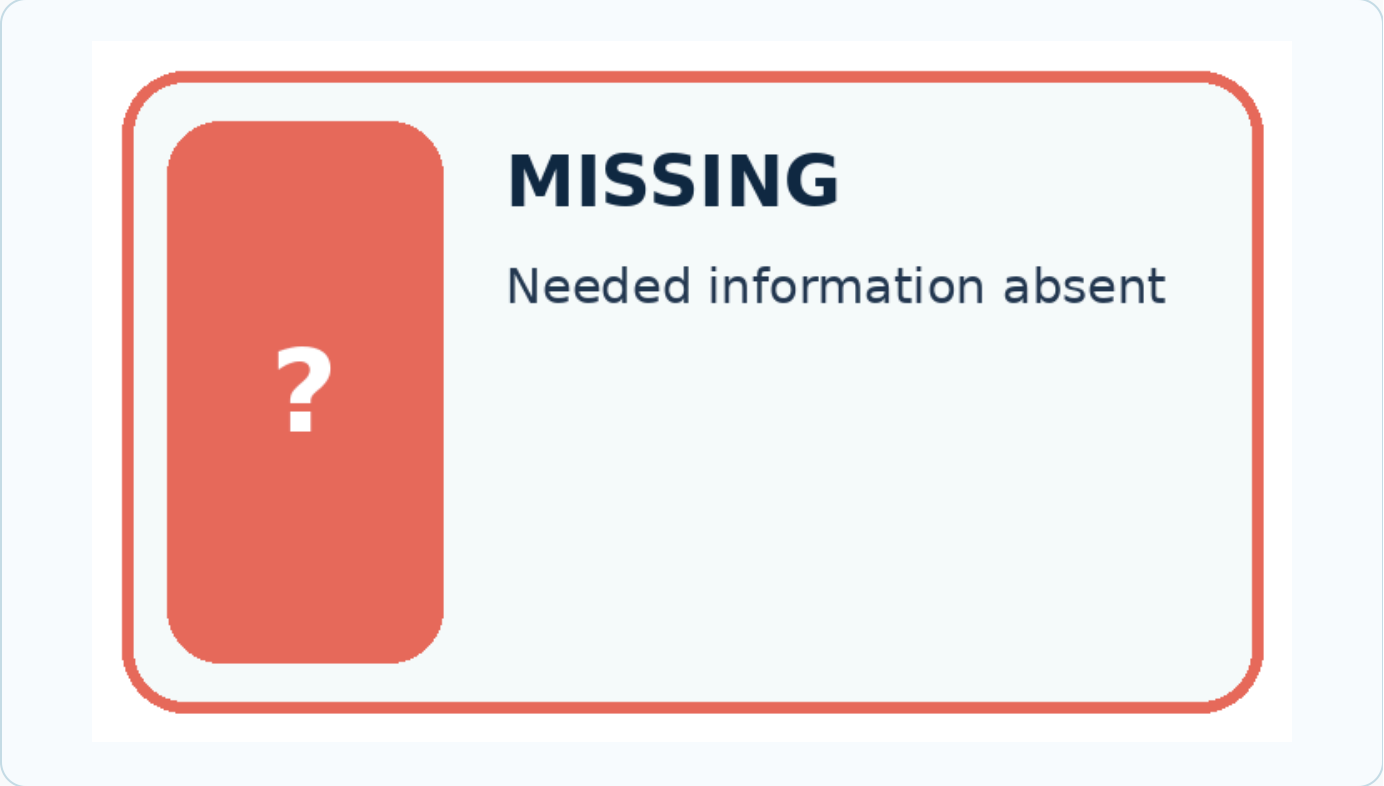
Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Extend the duration cases, then use a separate stock contract

> Duration maximum60 and stock maximum 8 belong to different tasks.

Predict the two added duration cases

Add {"minutes": 60} and {"minutes": "5"}. Predict the accepted list and rejection count.

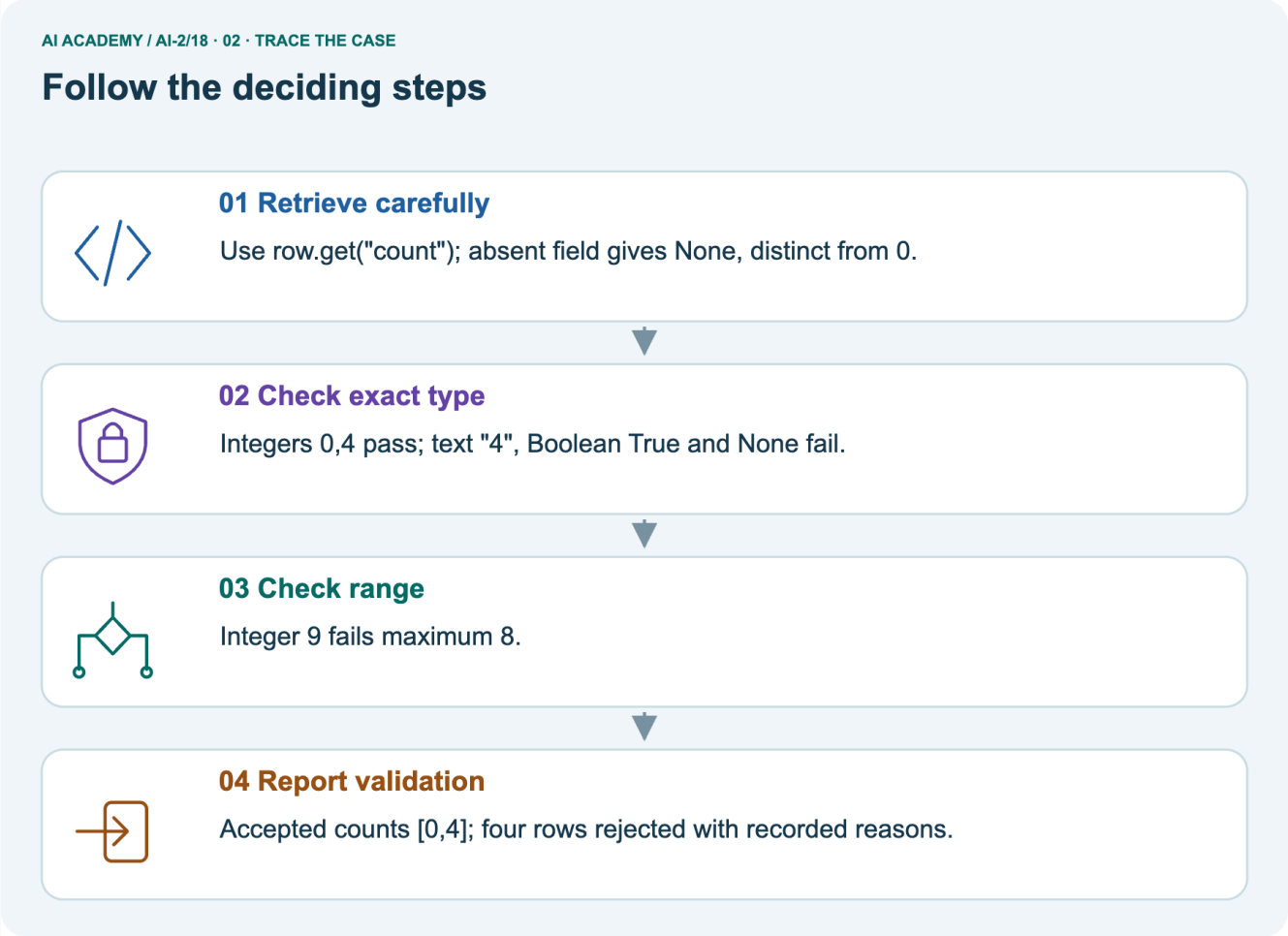
State the separate stock-count rule

A fictional catalogue records counts of available copies under a strict **input** contract.

Accept count only if type(value) is int and 0<=value<=8. Missing fields are invalid, and Booleans are not counts.

Rows have count 0, count 4, absent count, count "4", count True and count 9.

Trace the pictured exact-type and range checks



Follow the labelled picture with the matching explanation. Open the vector diagram

Testing if value would wrongly discard zero, which is a real valid count. Python's isinstance(True,int) is true, so the stated exact-integer contract uses type(value) is int to reject Boolean **input**.

Keep missing, type-invalid and range-invalid reasons separate rather than turning every rejected value into zero.

Run the complete stock validator and retain reasons

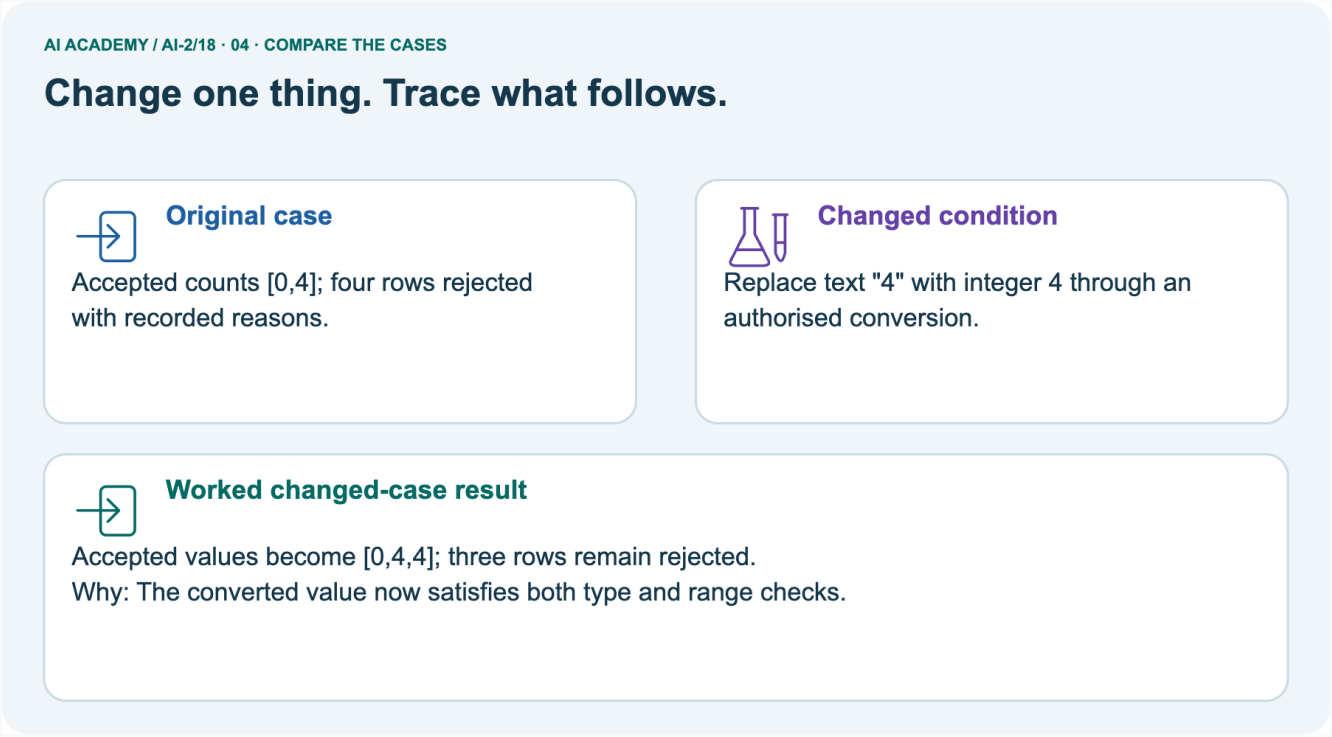
```
rows = [{"count": 0}, {"count": 4}, {},
        {"count": "4"}, {"count": True}, {"count": 9}]
accepted = []
```

```
review = []
for number, row in enumerate(rows, 1):
    value = row.get("count")
    if value is None:
        reason = "unknown count"
    elif type(value) is not int:
        reason = "count is not an exact integer"
    elif not 0 <= value <= 8:
        reason = "count outside 0 through 8"
    else:
        accepted.append(value)
        continue
    review.append((number, row.copy(), reason))
print(accepted)
print(len(review))
```

Predict one authorised text conversion

- Change one thing: Replace text "4" with integer 4 through an authorised conversion. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain Boolean rejection and the truth limit

- Read the labelled picture: Why is the Boolean True rejected even though arithmetic can treat it like 1 in Python?
- Type and range checks cannot confirm whether the real shelf count was measured accurately.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

WEEK 18 • WORKED EVIDENCE

1. Reader’s length preference is missing.
2. Ask: best when the user can answer.
3. Abstain: safest if the field is essential; default: convenient but may misrepresent.
4. Record which policy was used so the explanation stays truthful.

1. Reader’s length preference is missing.

2. Ask: best when the user can answer.

3. **Abstain**: safest if the field is essential; **default**: convenient but may misrepresent.

4. Record which policy was used so the explanation stays truthful.

Recalculate, retrace, or restate the reasoning in your own representation:

> **Write or draw your reasoning and evidence.**

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

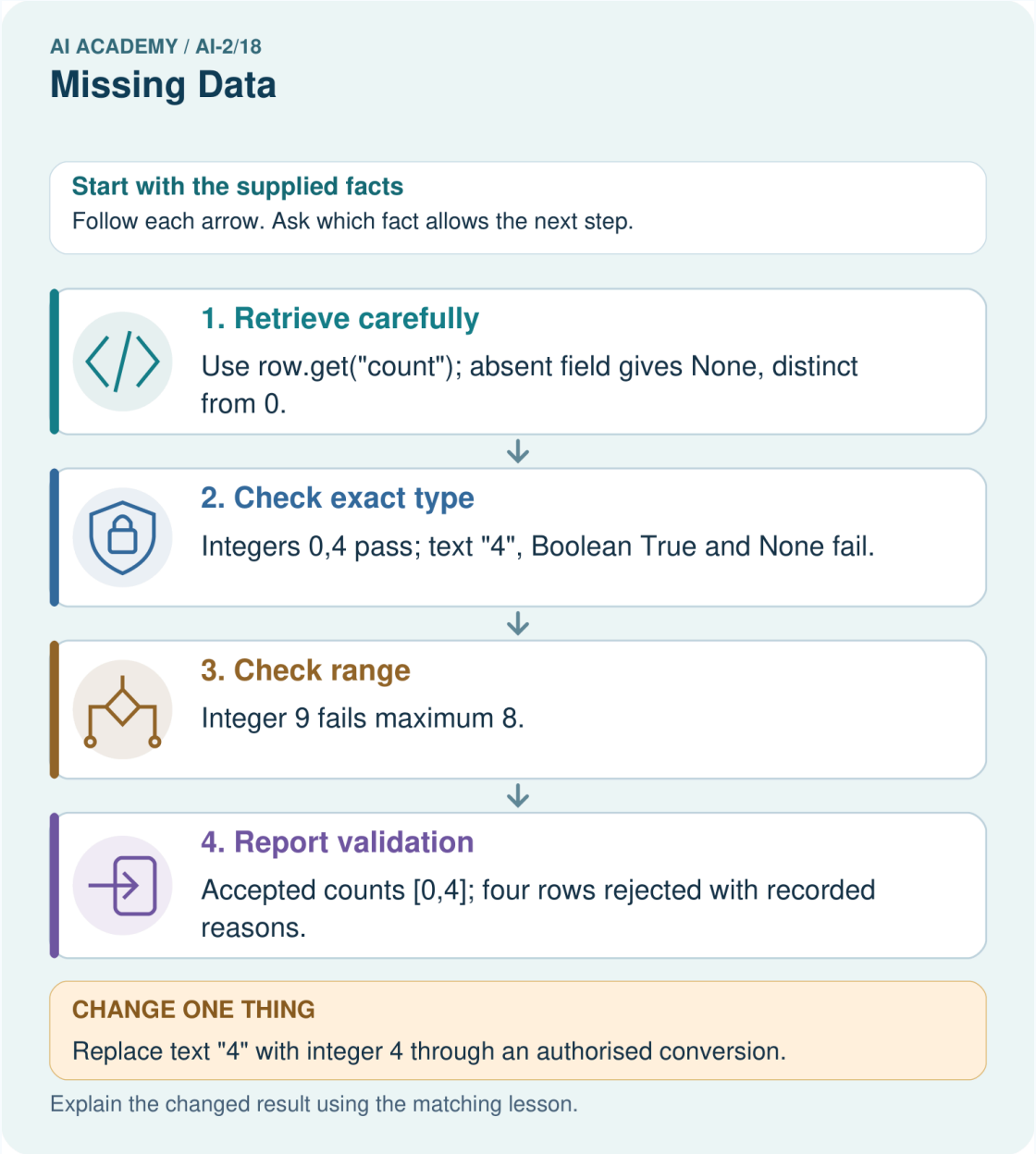
A closer explanation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Missing Data

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Retrieve carefully	Use row.get("count"); absent field gives None, distinct from 0.
2. Check exact type	Integers 0,4 pass; text "4", Boolean True and None fail.
3. Check range	Integer 9 fails maximum 8.
4. Report validation	Accepted counts [0,4]; four rows rejected with recorded reasons.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Report what filtering leaves out

> A statistic on accepted rows does not describe every supplied session.

Keep the five-row coverage statement

Rejection can change which sessions remain represented. Report 2 accepted of 5 supplied here. A summary of accepted rows does not automatically describe the missing or rejected sessions.

Explain why2/5 belongs in the report

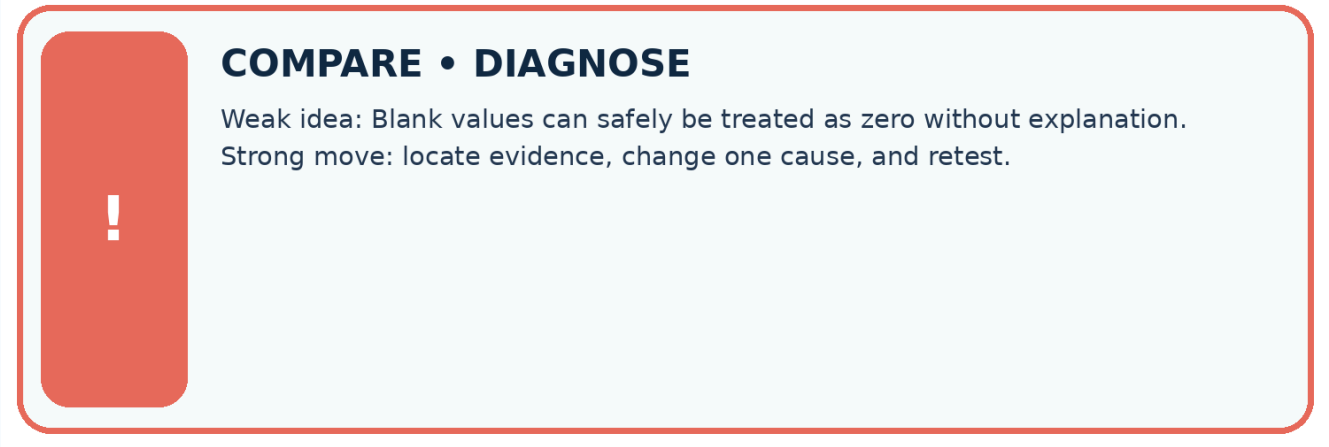
> **Only 2 of 5 supplied durations were accepted. Why should the report include this?**

▼ Earlier wording, alternatives and picture transcripts

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.



Evidence record	Expected	Actual	Likely cause / next test
normal case	works		

Evidence record	Expected	Actual	Likely cause / next test
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

> **Write or draw your reasoning and evidence.**

A closer explanation

MAIN LESSON · SECTION 8

Replace if value with the explicit contract

> Truthiness cannot distinguish allowed zero from invalid text.

Expose both sides of the shortcut error

Using if value would discard the legitimate zero and accept some invalid nonempty text. Test the written type and range rules explicitly. Filling an absent duration with 0 falsely claims an observed zero.

Trace the standalone W2 cases

Use workbook W2 for the connected example "**Missing value**s and **validation**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Challenge successful run proves the conclusion

- Use the worked case for **Missing value**s and **validation**. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the separate stock Boolean case

Why is the Boolean True rejected even though arithmetic can treat it like 1 in Python?

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.

TRACE THE LOGIC

Predict the result. Trace every condition or calculation. Compare expected with actual. Explain the first difference.

SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Practise with less help: Trace filtering

Check your reasoning

Explain the picture

My explanation using a pictured step and the given facts

MAIN LESSON · SECTION 9

Audit eight supplied records and preserve every review reason

> Distinguish baseline coverage from the expanded investigation.

The complete extended duration program

```
rows = [{"minutes": 0}, {}, {"minutes": 12},
        {"minutes": -1}, {"minutes": None},
        {"minutes": 60}, {"minutes": "5"}, {"minutes": True}]
accepted = []
review = []
for number, row in enumerate(rows, 1):
    value = row.get("minutes")
    if value is None:
        reason = "unknown duration"
    elif type(value) is not int:
        reason = "duration is not an exact integer"
    elif not 0 <= value <= 60:
        reason = "duration outside 0 through 60"
    else:
        accepted.append(value)
        continue
    review.append((number, row.copy(), reason))
print(accepted)
print(len(review))
```

Predict, trace, inspect and report coverage

1. Use the extended eight-record duration program supplied here. Before executing, record raw value, acceptance/review decision and first failing reason for each numbered row. The original five records accept2/5; adding60 and text"5" gives3/7; adding Boolean True gives3/8. Preserve those populations separately. Predict the accepted list and review count, then execute or label a paper trace. Inspect review to see retained row copies and reasons: unknown, non-integer or out of range. continue moves to the next row after a valid append; a rejected row reaches review.append. No source record is overwritten and no missing value is invented. Reserve the 0–10 independent count filter for W4.

Keep the evidence and explain the limit

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** fresh transfer
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

> **Write or draw your reasoning and evidence.**

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Optional policy extension · Four supplied scenarios

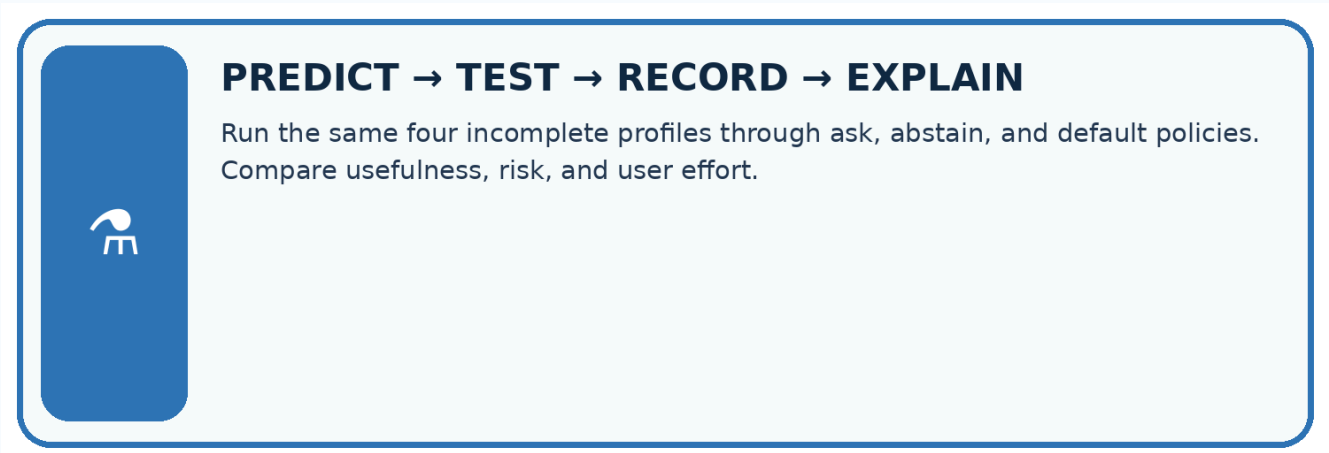
Optional recommender policy extension, using fictional scenarios: P1 has an unrecorded length preference; P2 has a missing page count required by a hard limit; P3 has unknown stock; P4 has a reader-confirmed no preference. Compare asking for missing evidence, abstaining from the decision that needs it, and a declared configured fallback. State what remains unknown and any user effort; do not invent a reader reply or present a default as a measurement. P4 is a known preference statement, not missing data. These policy choices are separate from the numeric filter and cannot change an unknown measurement into observed zero.

▼ Earlier wording, alternatives and picture transcripts

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Use the supplied extended eight-record duration audit for the main investigation. Retain accepted values and rejected row copies with reasons. The four explicitly supplied fictional recommender scenarios are an optional ask/**abstain**/fallback extension, not missing activity-log measurements.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: implement the new 0–10 count contract

> Write the new filter before consulting the key.

The independent row list and question

Write a filter for count values that must be integers from 0 to 10. **Process** {count: 3}, {}, {count: 0}, {count: 11}, {count: "3"}, using quoted keys in Python. Report accepted values and rejected count.

The exact-type and bounds policy

For this independent filter require exact Python int values, excluding Booleans, and inclusive bounds 0–10. Keep source order. Missing values are reviewed rather than imputed. The prompt's row keys must be quoted when writing executable Python; this task uses a different field and maximum from the worked duration and stock cases.

Your first attempt and support record

> **My independent explanation, trace and limits**
>
> ---
>
> ---
>
> ---
>
> ---

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after the attempt

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Validate independently

Week 18 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A book's page count is blank. Is blank the same as zero? Give an honest way for a short-book recommender to respond.

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Use ‘no preference’ only when the reader confirms it; otherwise ask or explain uncertainty.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

Scores award 2 for preferred genre and 1 for short length. A book matches genre but page count is missing. Compare rejecting it, assigning zero length points and asking for pages.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Keep numeric validation separate from preference policy

> Defaults and confirmed preferences need honest labels.

This week’s concrete portfolio contribution

For the fictional recommender, keep stock 0, missing stock and a reader-confirmed no preference as different states. Validate numeric fields against their own contract; do not silently convert text or fill unknown values. A partial score must disclose unknown components rather than claim a complete score. Record accepted and reviewed case counts, reasons, the chosen policy and the human next step. Use fictional profiles; do not ask real learners for extra personal information for this exercise.

Validate recommender stock and numeric features explicitly, recording zero stock as measured zero rather than missing.

Keep the checkpoint and human choices visible

THIS WEEK’S CHECKPOINT Use ‘no preference’ only when the reader confirms it; otherwise ask or explain uncertainty.

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits.
Never confuse a ranking with a fact about a person.

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



Evidence for your project

MAIN LESSON · SECTION 16

Finish: explain a valid zero and an unavailable observation

> State the contract, review reason and remaining coverage limit.

Explain without notes

- Close your notes. Explain this goal in your own words: Detect absent and unusable values

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	My own example	Not this / boundary
missing data	Information that was not recorded or is unavailable, which is different from an observed zero.	
default	The value or behaviour used when no alternative is specified.	
abstain	Decline to make a prediction when the available evidence does not meet the decision rule.	
impute	Replace a missing value using a stated method; an imputed value is an estimate, not an observation.	

Terms for the connected explanation

Word	Meaning
Missing value	A value that was not recorded

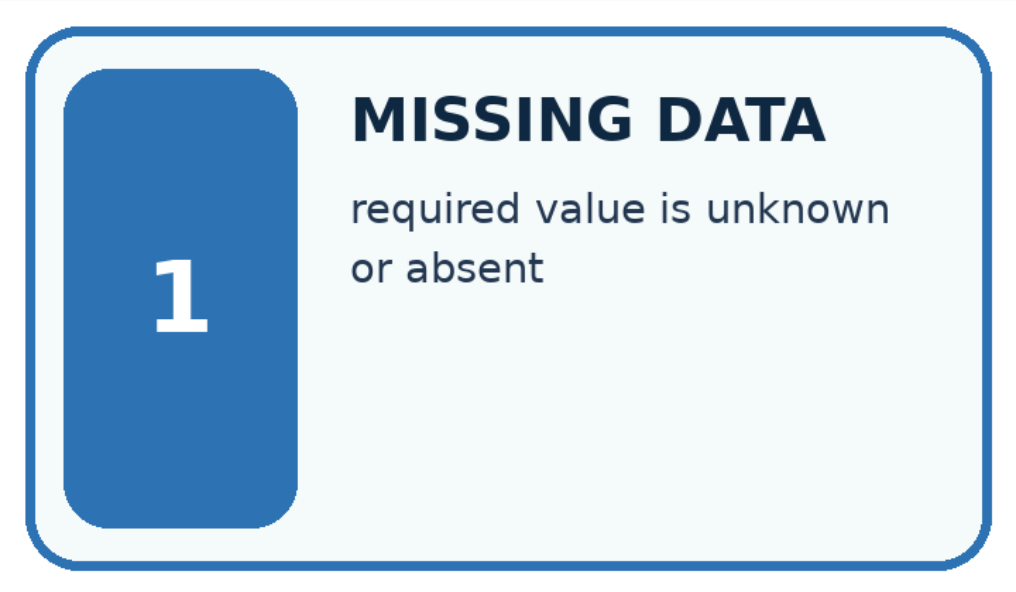
Word	Meaning
Validation	Checking data against explicit rules
Rejection	Holding a record out because it fails a rule
Coverage	Which intended records or groups are represented

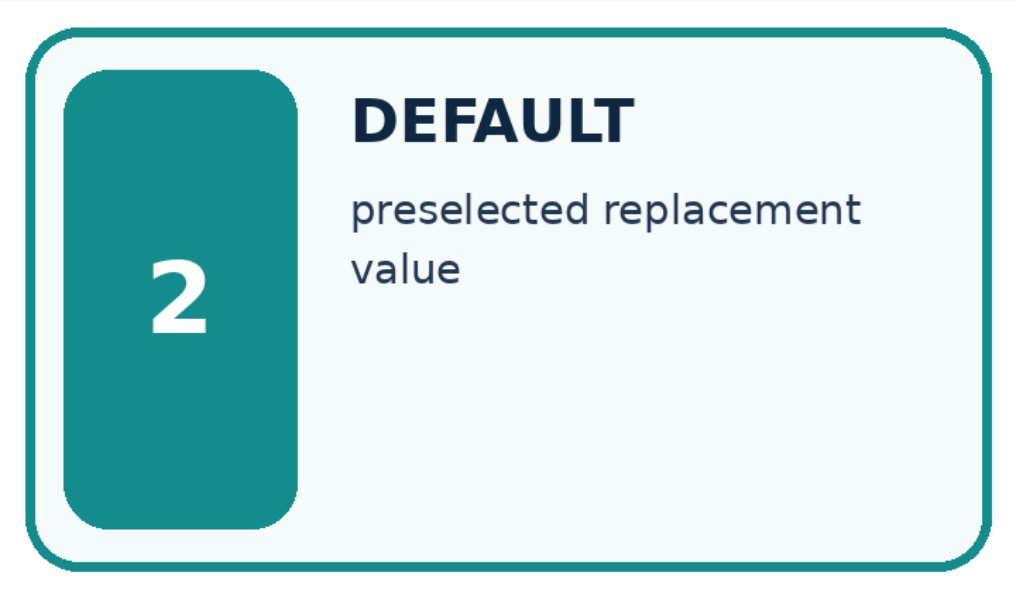
▼ Earlier wording, alternatives and picture transcripts

VOCABULARY LAB

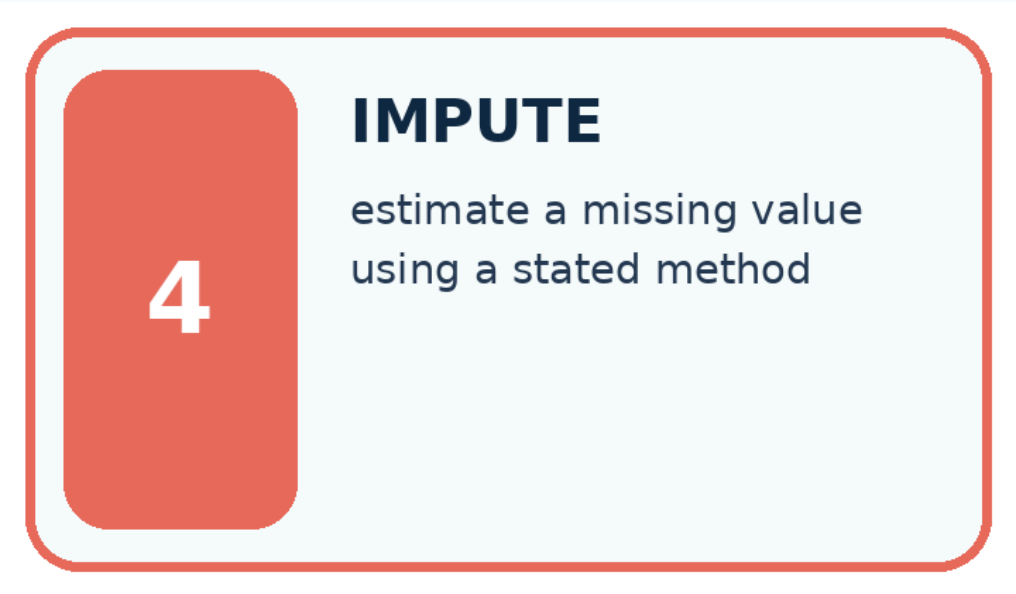
Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.




2
DEFAULT
preselected replacement value


3
ABSTAIN
decline to decide



Words used in the closer explanation

▼ **Choose a stronger thinking challenge**

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

- My strongest **evidence** of independent understanding:

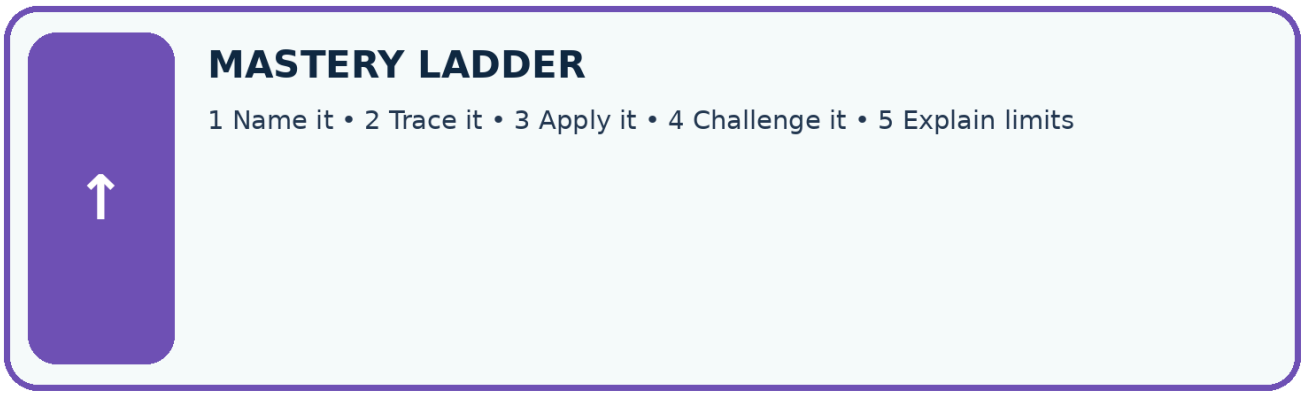
Write or draw your reasoning and evidence.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference: check your explanation

Reference: check your explanation

> Use after practice. Validate independently is the selected independent case.

Check the evidence and its limit

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One limitation, fairness concern, or situation needing human review:

> **Write or draw your reasoning and evidence.**

> **Write or draw your reasoning and evidence.**

Self-check and record

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

▼ **Earlier wording, alternatives and picture transcripts**

INDEPENDENT MASTERY PROOF

Checklist for explaining this week's idea

Use after practice. Validate independently is the selected independent case.



EXPLANATION CHECK Use an exact type and explicit bounds, preserve zero, retain review reasons and report the population that remains.

Write or draw your reasoning and evidence.

▼ **Extra practice: Missing is different from zero**

REFERENCE · SECTION 13

Extra practice: Missing is different from zero

> Try the question before opening its worked explanation.

Read the supplied case and question

Genre is known, but page count is missing and length affects recommendation score.

Choose transparent handling.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

Retrieve or ask for the count, or use a documented partial-score policy with uncertainty. Do not silently insert zero.

Evidence or stage	Value or result
Retrieve	Look up the page count
Ask	Request the missing information
Partial score	Use a documented policy with uncertainty

▼ Earlier wording, alternatives and picture transcripts

Week 18 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Missing is different from zero

Genre is known, but page count is missing and length affects recommendation score.

Choose transparent handling.

Retrieve or ask for the count, or use a documented partial-score policy with uncertainty. Do not silently insert zero.

▼ **Reference: original lesson checklist**

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Detect absent and unusable values	Explain the deciding step, show a trace or test, and state one limit.
Keep unknown distinct from zero	Explain the deciding step, show a trace or test, and state one limit.
Report accepted and rejected records	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: A fictional catalogue records counts of available copies under a strict **input** contract.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional catalogue records counts of available copies under a strict **input** contract.
- Which labelled step matches this detail? Accepted counts [0,4]; four rows rejected with recorded reasons.
- 1. Report **validation**
- 2. Retrieve carefully
- 3. Check range

- 4. Check exact type

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Report **validation** Accepted counts [0,4]; four rows rejected with recorded reasons. Testing if value would wrongly discard zero, which is a real valid count.

Python's isinstance(True,int) is true, so the stated exact-integer contract uses type(value) is int to reject Boolean **input**.

Keep missing, type-invalid and range-invalid reasons separate rather than turning every rejected value into zero.

Changed-case question

- Changed-case practice: Replace text "4" with integer 4 through an authorised conversion. Which conclusion is supported by the supplied facts?

- 1. We can decide the result without examining the changed **input** or the procedure.

- 2. This one example guarantees correct results for every future **input**.

- 3. Accepted values become [0,4,4]; three rows remain rejected.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> ---
>
> ---
>
> ---
>
> ---

▼ **Check the changed case**

After your attempt, compare with the supported explanation: Accepted values become [0,4,4]; three rows remain rejected. The converted value now satisfies both type and range checks.

Repair and save your work

Review Noise and Outliers (ai-2/17). Goal: Calculate mean and median. Short practice: If 18 was genuine, which summary better describes the middle observed task, and what extra fact should be shown?.

Return to this week and explain what you changed.

Review Choose Informative Questions (ai-2/15). Goal: Fit a root question and leaf **prediction**s from **training** examples. Short practice: Choose the best original question and explain why its more balanced split matters here..

Return to this week and explain what you changed.

Review Accuracy as a Fraction (ai-2/12). Goal: Explain why overall accuracy can hide failure on a less common class. Short practice: Why is 100% repair detection not sufficient to declare the changed classifier best?.

Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work
</details>

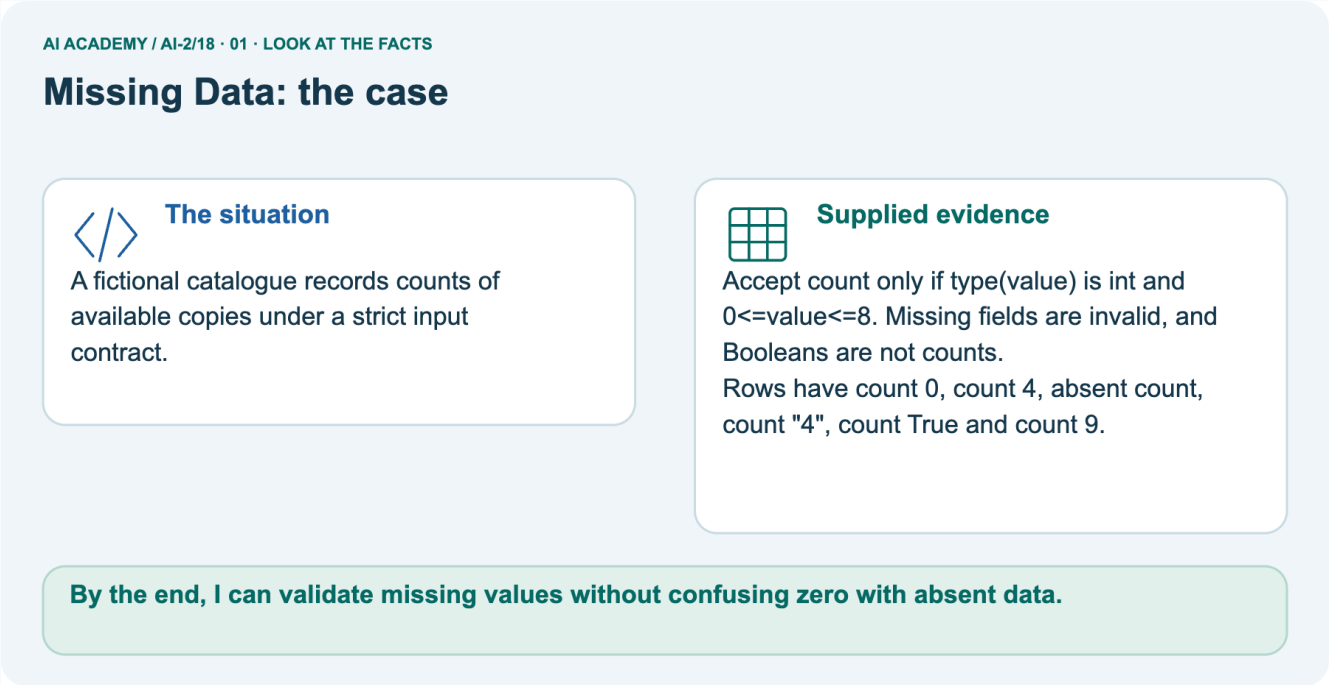
<details>
<summary>Reference: the six-picture journey</summary>
</details>

REFERENCE · SECTION 18

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Missing Data: the case



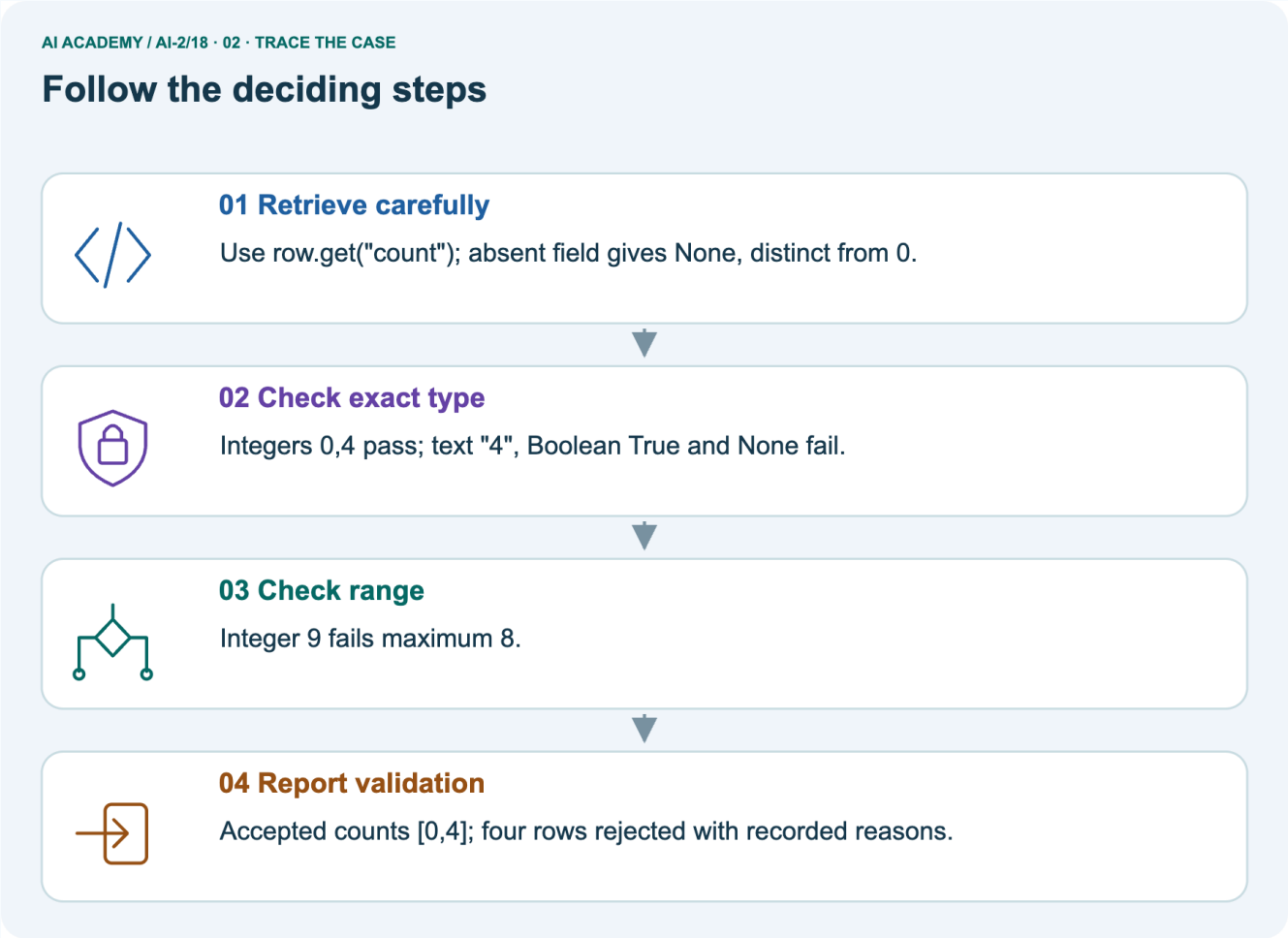
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional catalogue records counts of available copies under a strict **input** contract.
Supplied facts: Accept count only if type(value) is int and 0<=value<=8. Missing fields are invalid, and Booleans are not counts.

Rows have count 0, count 4, absent count, count "4", count True and count 9.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

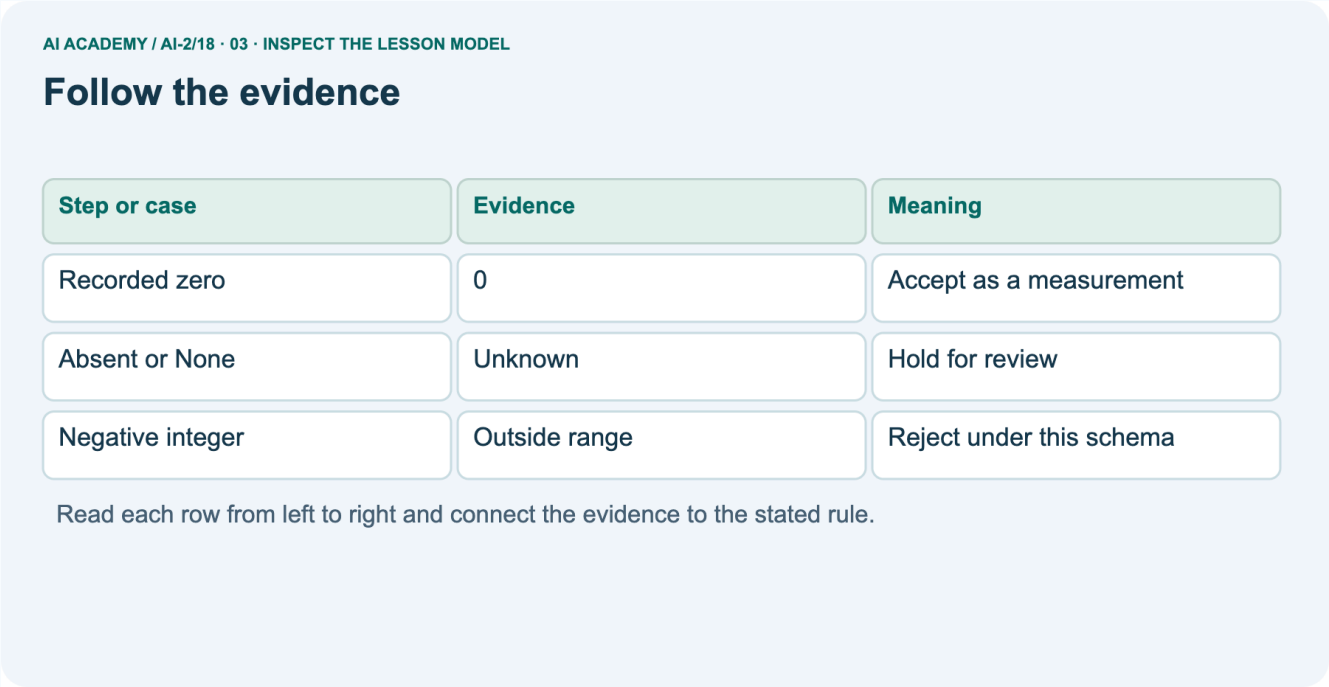


Follow the labelled picture with the matching explanation. Open the vector diagram

Retrieve carefully: Use row.get("count"); absent field gives None, distinct from 0. → Check exact type: Integers 0,4 pass; text "4", Boolean True and None fail. → Check range: Integer 9 fails maximum 8. → Report **validation**: Accepted counts [0,4]; four rows rejected with recorded reasons.

Point to each numbered stage and explain which recorded **input** it uses.

03. Follow the evidence

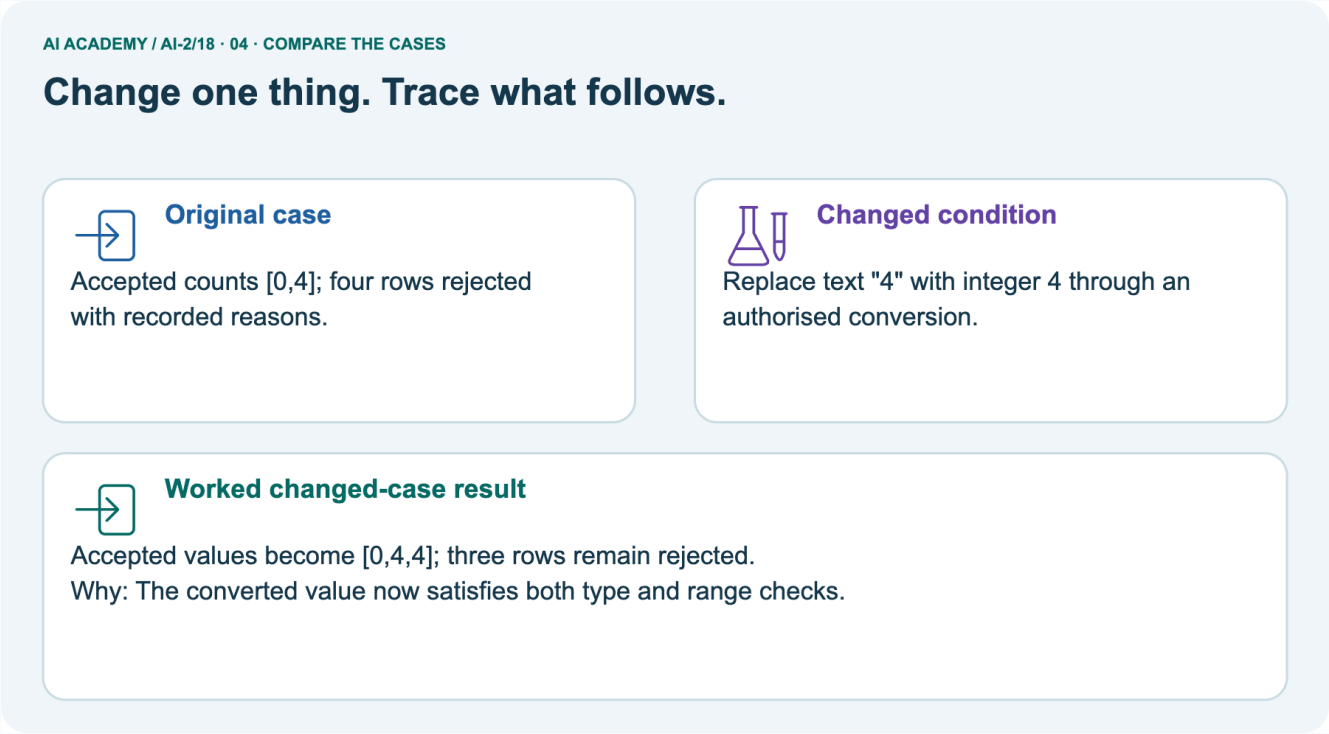


Follow the labelled picture with the matching explanation. Open the vector diagram

Follow the **evidence**. Recorded zero | 0 | Accept as a measurement. Absent or None | Unknown | Hold for review. Negative integer | Outside range | Reject under this schema

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



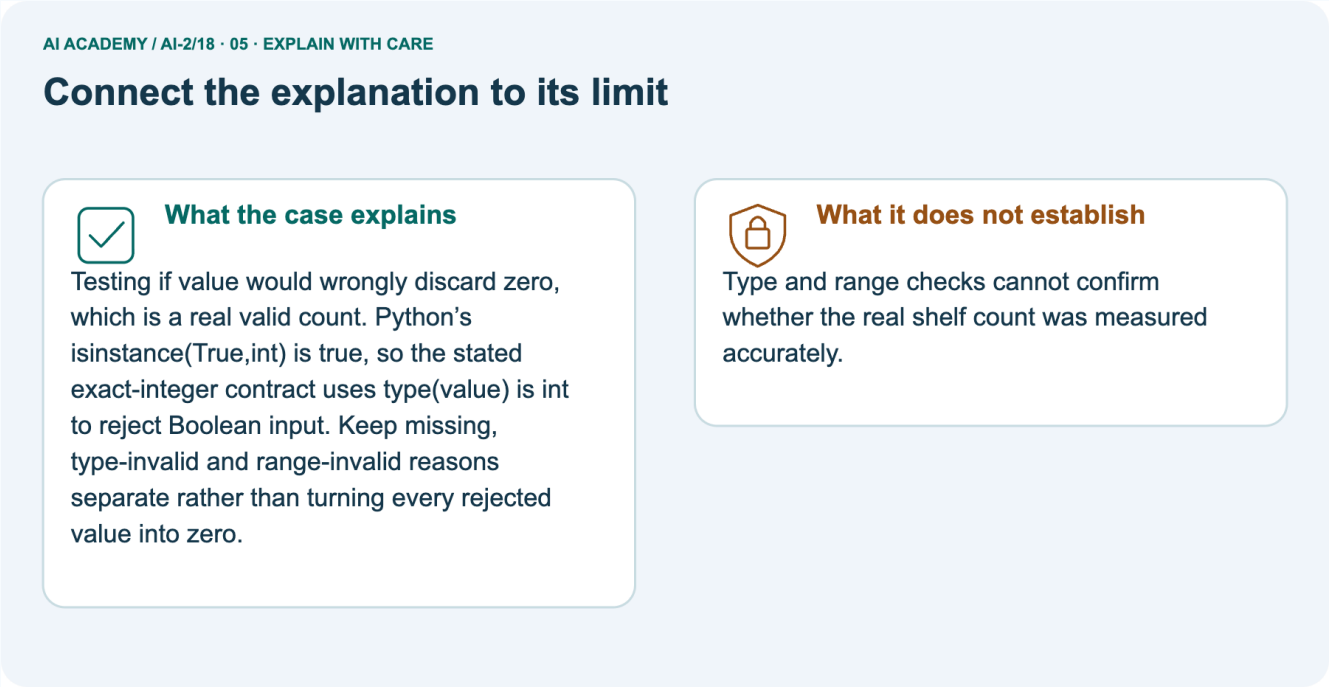
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Accepted counts [0,4]; four rows rejected with recorded reasons. Changed condition: Replace text "4" with integer 4 through an authorised conversion.

Expected result: Accepted values become [0,4,4]; three rows remain rejected. Reason: The converted value now satisfies both type and range checks.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



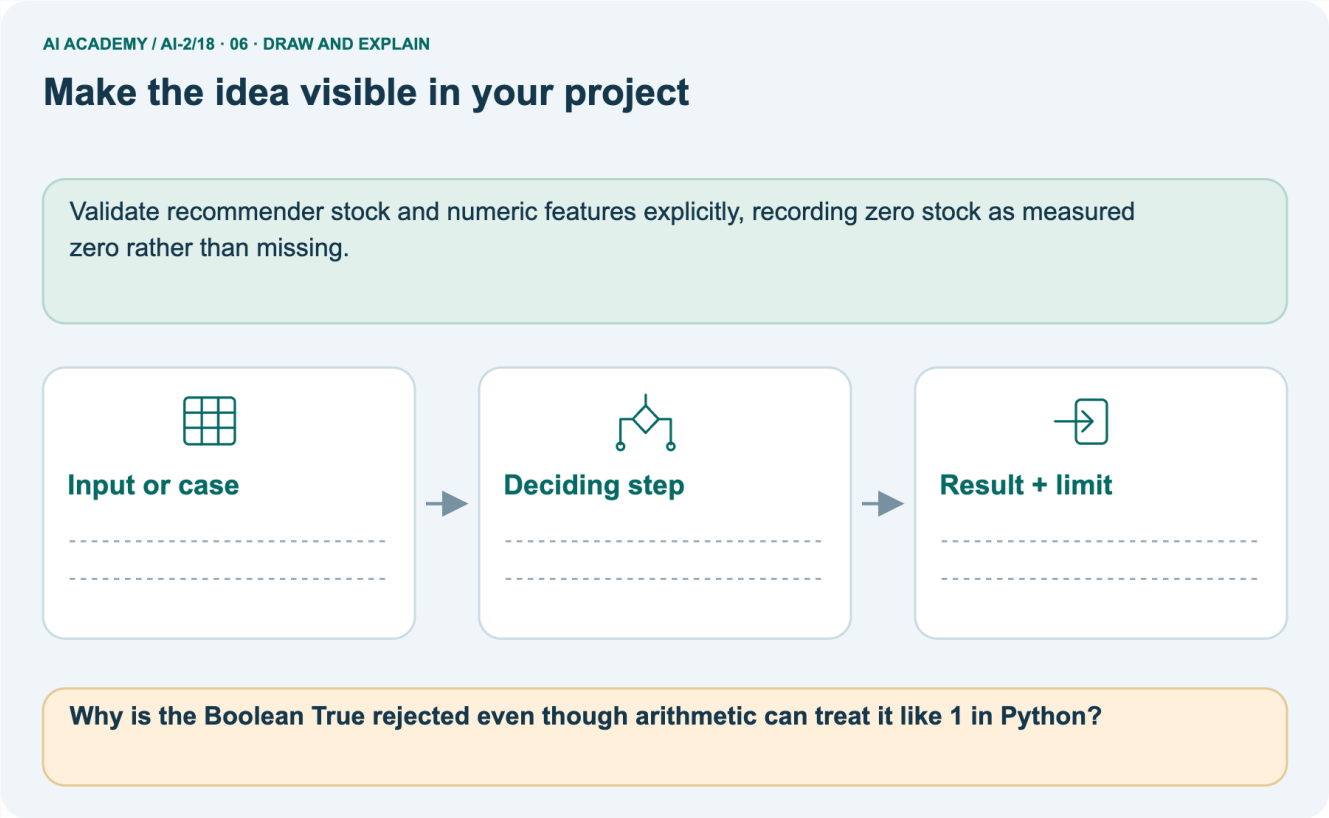
Follow the labelled picture with the matching explanation. Open the vector diagram

Testing if value would wrongly discard zero, which is a real valid count. Python’s isinstance(True,int) is true, so the stated exact-integer contract uses type(value) is int to reject Boolean **input**.

Keep missing, type-invalid and range-invalid reasons separate rather than turning every rejected value into zero. Limit: Type and range checks cannot confirm whether the real shelf count was measured accurately.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Validate recommender stock and numeric features explicitly, recording zero stock as measured zero rather than missing. Draw your **input**, deciding step, result and limit.

Supported question: Why is the Boolean True rejected even though arithmetic can treat it like 1 in Python?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Missing Data: the case

02. Follow the deciding steps

03. Follow the evidence

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.